Completed results: partial experimental agreement

The latest source-visible concrete/PDROD comparison completed 3 ms: force peak 298.952 kN, -0.68% from the cited experiment. Fixed bottom motion is 27.570 mm, +17.92% from its reported maximum and still descending. This is partial agreement, not an exact Bowen reproduction.

Report generated 2026-10-13 14:28:31 UTC+08. All listed computational windows completed. The latest frozen model and results are preserved without further parameter changes, as requested. Event maxima, experimental sensor equivalence and punching/shear crack agreement remain unvalidated.

Case / evidence	Peak force [kN]	Probe [mm]	Window [ms]
Bowen, Table 3-1	359.800	22.700	Published maxima
Experiment cited by Bowen	301.000	23.380	Published maxima
Current-stretch concrete + PDROD	298.952	27.570	3.000
Current-stretch PDROD only	272.977	33.668	3.000
Dynamic-modulus diagnostic	132.463	19.789	3.000
Static-c reconstruction, dt 0.30 µs	272.977	253.613	20.0001
Static-c reconstruction, dt 0.15 µs	278.938	33.218	3.000
Static-c reconstruction, dt 0.075 µs	293.642	2.213	.300
Static-c reconstruction, dt 0.0375 µs	362.702	2.213	.300
Recovered original model	220.663	38.089	3.000

Computed probes are finite-window motion of fixed concrete particles, not validated maximum deflections. The latest 3 ms model retains all bottom probes and steel nodes bearing-connected. The older 20 ms branch loses both load paths; its 253.613 mm value is retained as failed-candidate evidence.

Raw force maxima remain sensitive to the nodal contact operator. The finest-step 362.70 kN peak is a verified activation spike. No temporal/mesh convergence is asserted for the latest history-policy comparison.

No material input, artificial damping, contact gap or force filter was fitted. Source-visible history-policy comparisons remain distinct from recovered author inputs. A separately predeclared dynamic-modulus candidate is assessed in the addendum. An isolated source-visible PDROD force-history comparison is also retained. A further isolated concrete force-history comparison is included.

Published inputs and equation-to-code audit

Quantity	Reconstructed input
Beam / support span	600 × 70 × 100 mm / 500 mm
Particle spacing / horizon	5 mm / 15 mm; 33,600 concrete particles
Concrete	E 25 GPa; ft 4.6 MPa; fc 60 MPa; ρ 2400 kg/m³
Fracture energy / steel	105 J/m²; E 200 GPa; fy 482 MPa; ρ 7750 kg/m³
Longitudinal bars	Two 6 mm bars; centre 20 mm above beam bottom
Impactor	38.4 kg; 10.8 m/s; hemispherical tip radius 60 mm

Cell-centred concrete coordinates are combined with 270 steel nodes and 268 axial rods. Full concrete volume plus embedded steel mass is retained; concrete displaced by the bars is not subtracted in this declared hybrid representation.

Concrete force and Eq. 2-12 kernel

Each unique pair applies equal opposite physical forces c(r) × bond-law stretch × Vi Vj × volume/surface factors [N]. Micromodulus has units N/m⁶; acceleration divides by physical nodal mass.

PDROD steel–concrete transfer

122 virtual spherical samples per steel node use eight-corner interpolation. Weighted opposing concrete forces preserve net force and torque, and match interpolation virtual work. This proves conservation, not the missing interface strength/stiffness inputs.

Steel / Eqs. 2-45–47 contact

Corotational axial rods use elastic-perfectly-plastic return mapping. Sphere relocation changes nodal momentum; reaction is impulse/dt. The hammer receives equal recoil. Its horizontal guide impulse is recorded separately.

Damage / measurement

Eq. 2-10 cumulative failed-family volume is distinct from current force inactivity and crack opening. Connectivity excludes analytically zero horizon-shell interactions. Concrete, steel and hammer probes are reported separately.

References: thesis Chapters 2–3, PDROD Section 2.5 and Appendices A–C. Printed p. 32 / PDF p. 49 gives the target table. Contact is printed pp. 27–28 / PDF pp. 44–45. Fig. 3-4 is force versus deflection, not a recovered experimental time history.

Chapter 3 geometry: inputs and assumptions

The independent review re-extracts the actual PDF text, embedded images and vector paths. It recovers several current inputs directly; it does not identify a new source-proven geometry correction or change any simulation.

Definition	Source-supported value	Evidence
Beam / support span	600 × 70 × 100 / 500 mm	§3.1.1 and Fig. 3-1
Steel diameter / number	6 mm / two bars	Fig. 3-1 and §3.1.1
Steel level	80 mm from top → z = 20 mm	Dimensioned Fig. 3-1
Symmetric bearing axes	x = 50 and 550 mm	500 mm span, 600 mm beam
Hammer head	Hemisphere, radius 60 mm	Explicit Chinese word “radius”
Longitudinal supports	Left pin, right roller	Simply supported schematic

The 20 mm value is steel centreline height above the beam bottom. If the labeled 80 mm depth refers to the centreline, nominal clear bottom cover is 17 mm. Both Bowen and the supplied Wang paper state a 60 mm hammer radius; neither supports a radius/diameter correction.

Undimensioned vector drawing measurements give straight-bar x endpoints 20.000 and 579.988 mm and hook rise 59.996 mm. They narrow geometric inferences but do not prove solver coordinates. Transverse symbols do not provide reliable physical bar centres. The runs retain declared x = 25–575 mm, y = 20/50 mm and 60 mm hook rise. Fig. 3-2 confirms upward end hooks, without node spacing, hook length or bend-radius inputs.

Generic §2.6.4 specifies a virtual constraint region Rc of width δ, hence 15 mm here. Its figure shows an end-type boundary region. Chapter 3 does not map it to the bottom bearing x-footprint, z-depth or transverse extent. This evidence does not define a particular replacement for the current 10 mm bilateral bottom patches and minimal lateral anchors.

The pin/roller schematic supports the longitudinal simply supported interpretation. Bearing footprint, finite-spatial contact, lift-off, friction and complete 3D constraints remain missing. The current ID-based constrained particles later move 317 mm laterally; this is a limitation of that realization, not recovered physical bearing behavior.

Chapter 3 prescribes velocity at impact but not the initial hammer centre, clearance, or tangency to the physical surface versus point centres. With a 5 mm cell-centred mesh those top definitions differ by 2.5 mm. Source-defined contact processing and experimental probe equivalence remain unresolved.

Retained evidence: printed p. 30 / viewer p. 47, Fig. 3-1; printed p. 31 / viewer p. 48, Fig. 3-2; printed p. 26 / viewer p. 43, generic boundaries. Full excerpts, original images, vector coordinates, source hashes and audit script are preserved.

Corrections, scientific alternatives and unknowns

Physical units and PDROD action/reaction were checked directly. Contact reaction contains the physical mass and the new-time velocity change, exactly as printed in Eqs. 2-45–47. No remaining concrete-contact extra-volume, extra-dt, sign or velocity-reference error was found.

Fracture work: the printed terminal-stretch formula belongs to a constant kernel, while the selected micromodulus is weighted. Their combination is scientifically inconsistent. It is not a proven transcription error in the unavailable author executable.

$\Gamma = (\pi/2)\, s_0\, s_c \int_0^{\delta} c(r)\, r^4\, dr$
Constant kernel: $\Gamma = (6/5)\, E\, \delta\, s_0\, s_c$
Weighted kernel: $\Gamma = (32/35)\, E\, \delta\, s_0\, s_c$

Declared terminal definition	sc	Lattice mean Γ
Printed constant + weighted kernel	.001268115942	77.8259 J/m²
Weighted continuum work	.001664402174	102.1465 J/m²
Current weighted lattice mean	.001710898586	105.0000 J/m²

The continuum weighted/constant work ratio is 16/21. The terminal correction follows that integral and the independently calculated source lattice calibration; it does not use impact peaks. An orientation mean of 105 J/m² does not make all cleavage planes equal to 105.

Rate alternative: the printed join at 1/s drops the strain-threshold multiplier by 28.99%. Production uses a continuous join at 1/s with the printed branches. The printed coefficient is alternatively continuous at 3/s; Appendix A has another exponent. The original author’s chosen policy is unknown.

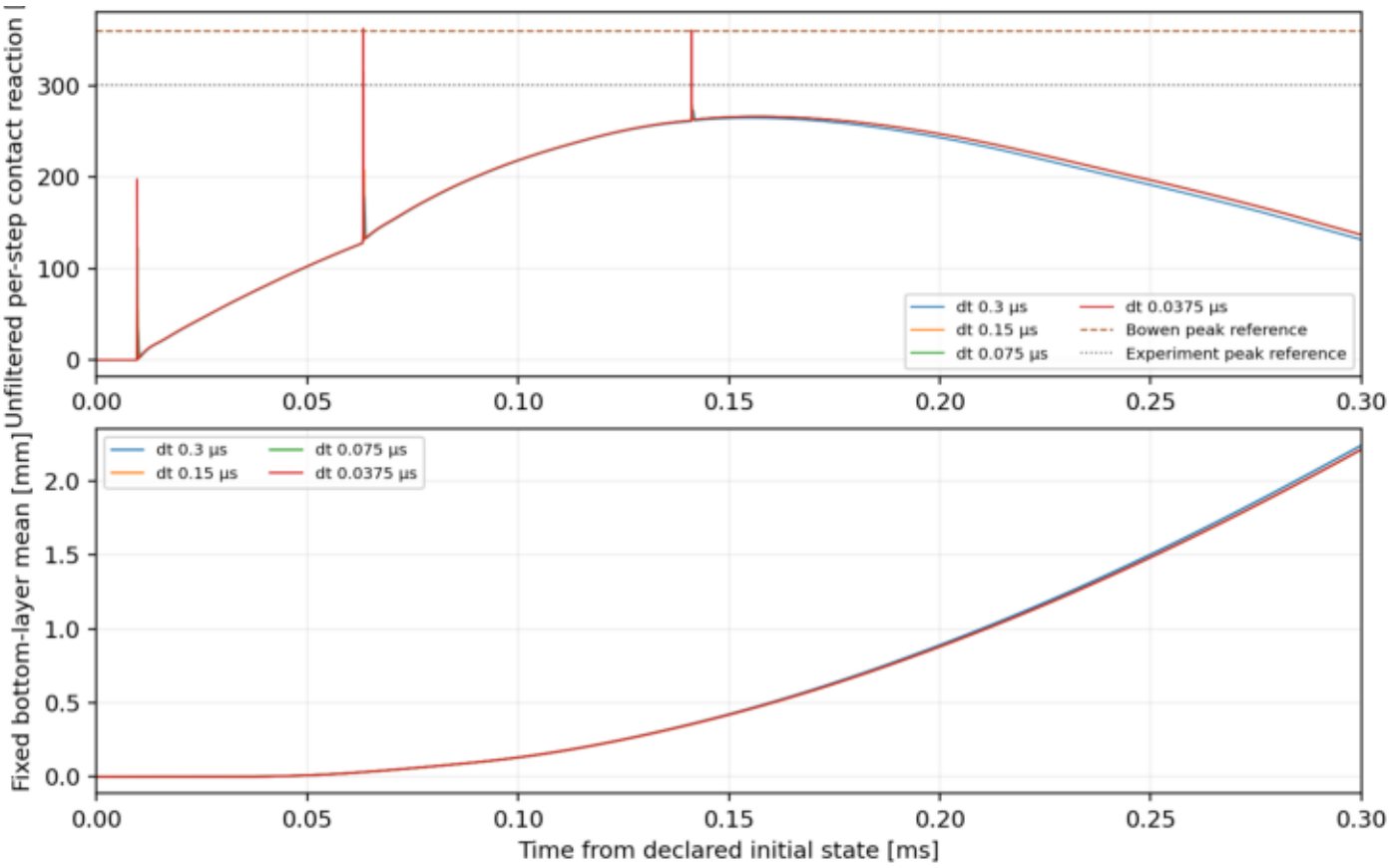
Reference-branch choices: irreversible history; static micromodulus; particle-top-tangent initial contact; concrete-weighted static interface law. No smoothing or contact stiffness was tuned. A separately declared dynamic-modulus candidate, when included, is assessed in the addendum.

Unrecovered definitions include interface bcc/s01/sc1/control volumes, bar ends and lateral positions, 60 mm hook rise, 10 mm bilateral bearing patches and anchors, original integration loop, hammer body, contact origin, rate history and sensor processing. The supplied different-case Fortran cannot identify these Chapter 3 inputs.

Isolated 3 ms interface-history sensitivity: ignoring previous failure revives 4,379 samples, including 1,174 in tension. The sum of nodal steel-force magnitudes changes from 8.09 kN to 3.413 MN; force-primitive potential changes from .015 to 1920 J. Both branches conserve force and torque. These are fixed-geometry evaluations. The visible Appendix C and different-case Fortran sample loops evaluate force before permanent bookkeeping, without consulting old failure flags. V3 gates failed samples; a separate fresh v4 comparison, when included, restores instantaneous PDROD evaluation. No mid-run switch is made.

Timestep refinement: raw peak does not converge

dt [µs]	Raw peak [kN]	Peak time [ms]	Impulse ≤.3 ms [N·s]
0.3	272.977271	0.1416000	54.731091
0.15	278.937795	0.1413000	55.420588
0.075	293.642356	0.1411500	55.438647
0.0375	362.701700	0.0632625	55.442454



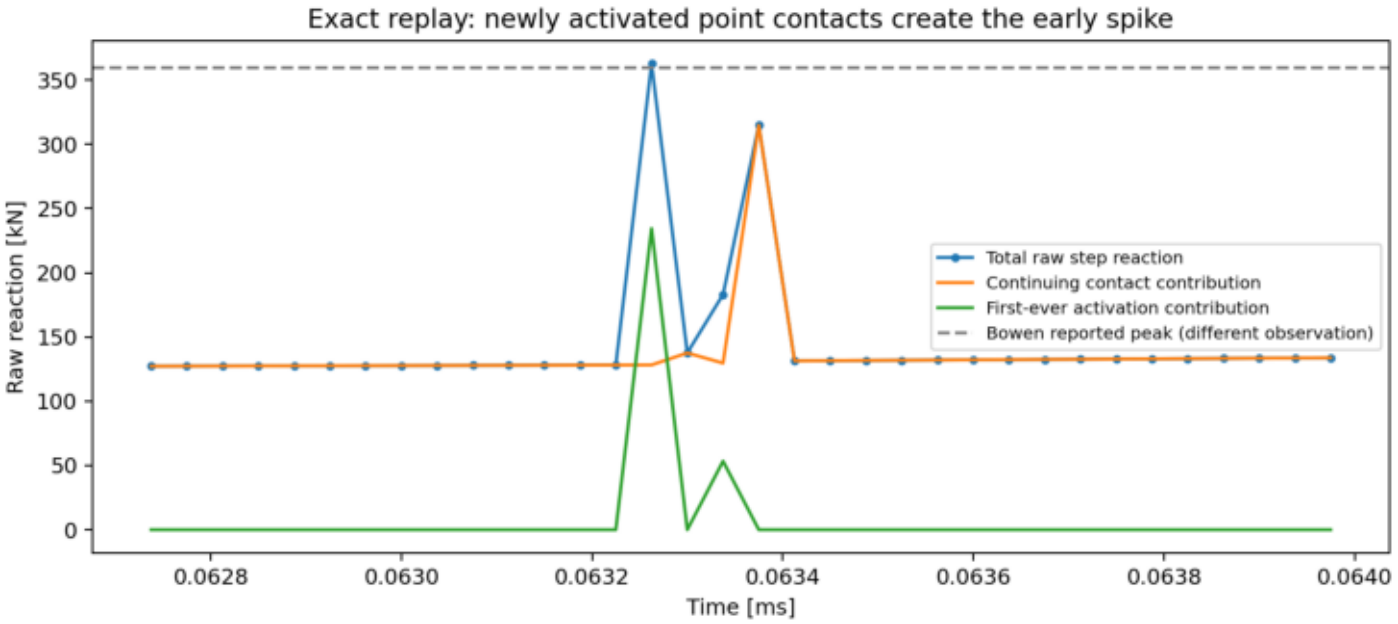
The raw peaks change by +2.18%, +5.27% and +23.52% under successive dt halving. At the finest dt the controlling peak moves from approximately 0.141 ms to 0.0632625 ms. These values do not support a finite extrapolated force maximum.

Coarse and half-dt share identical frozen v3 inputs except dt. At 3 ms, bottom motion is 33.906416 versus 33.217723 mm (-2.031%). Impulse is 65.085371 versus 66.431531 N·s.

Temporal sensitivity is separate from full impact mesh convergence, which has not been established. The nearly identical early displacement at the smallest steps does not repair the unfiltered force observation or the severe later structural response.

Verified contact activation spike: exact replay

All numeric fields of all 1,750 replayed rows match the retained original run exactly. The unchanged operator at step 1687, 0.0632625 ms gives 362.701700 kN. Four newly contacting nodes contribute 234.443369 kN; four continuing nodes contribute 128.258331 kN.



Each new point has mass 0.0003 kg. Projection changes its predicted vertical velocity from about −2.952887 to −10.279242 m/s. The four finite activation impulses total 0.008791626 N·s; dividing by dt = 3.75×10^{-8} s produces the large per-step sample.

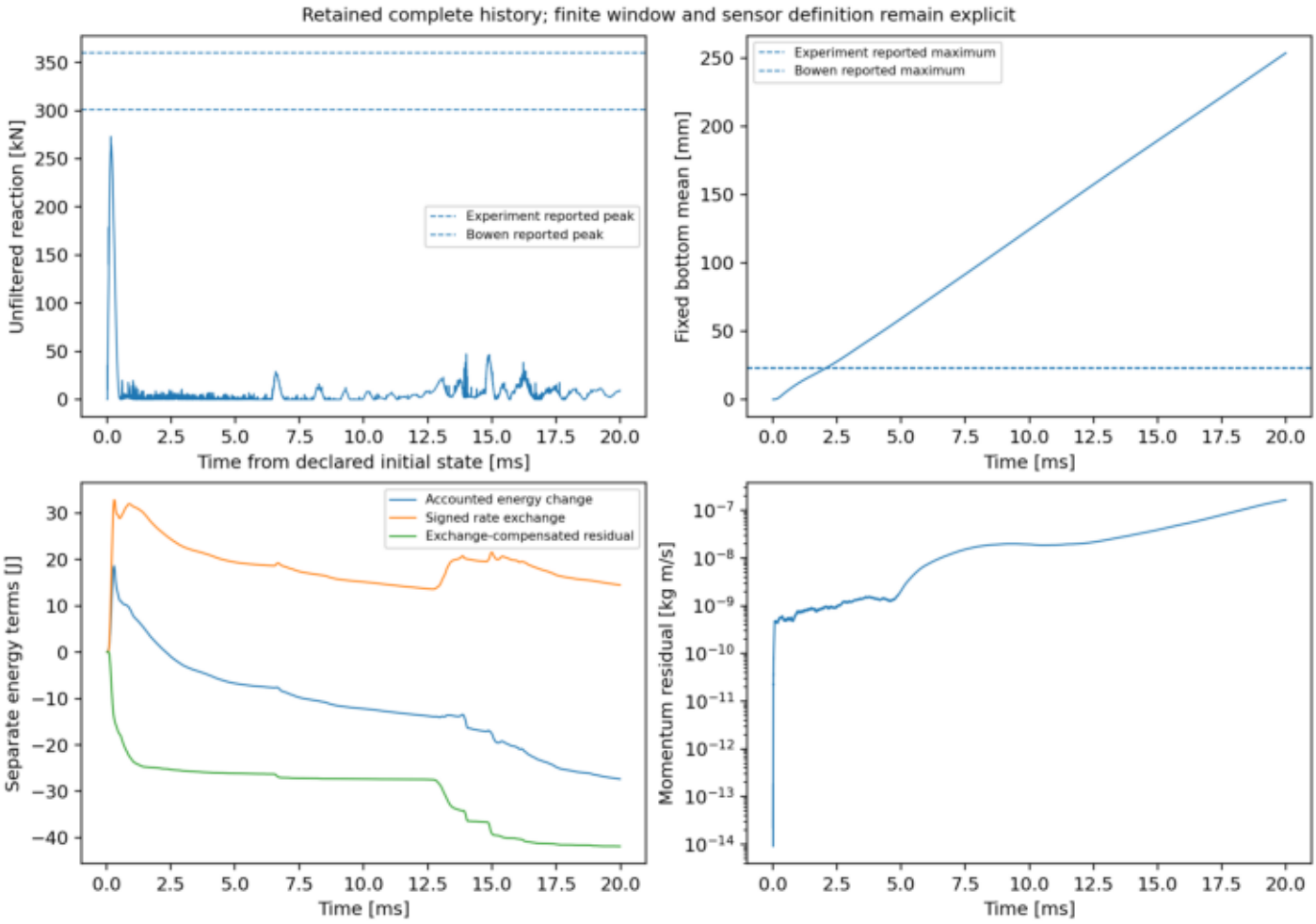
Actual-operator tangent-node verification: Δv = −V; I = mV; Fsample = mV/dt. Four resting points have total mass 0.0012 kg and incoming speed 10.8 m/s.

dt [μs]	Impulse [N·s]	Raw reaction [kN]
0.3	0.01296	43.2
0.15	0.01296	86.4
0.075	0.01296	172.8
0.0375	0.01296	345.6
0.01875	0.01296	691.2

An entry event may split its impulse over two bins. Time-grid phase, normals and pre-contact forces change individual peaks; they do not remove the finite nodal velocity jump. This is a rigid lumped point-contact limitation, not incorrect force normalization.

The experiment’s force bandwidth, sampling and peak-processing rule remain unknown. No supported filter or compliant-contact replacement was found. Refining until a spike approaches 359.8 kN is not reproduction, and no filter or contact parameter was selected to obtain agreement.

Completed 20 ms production response

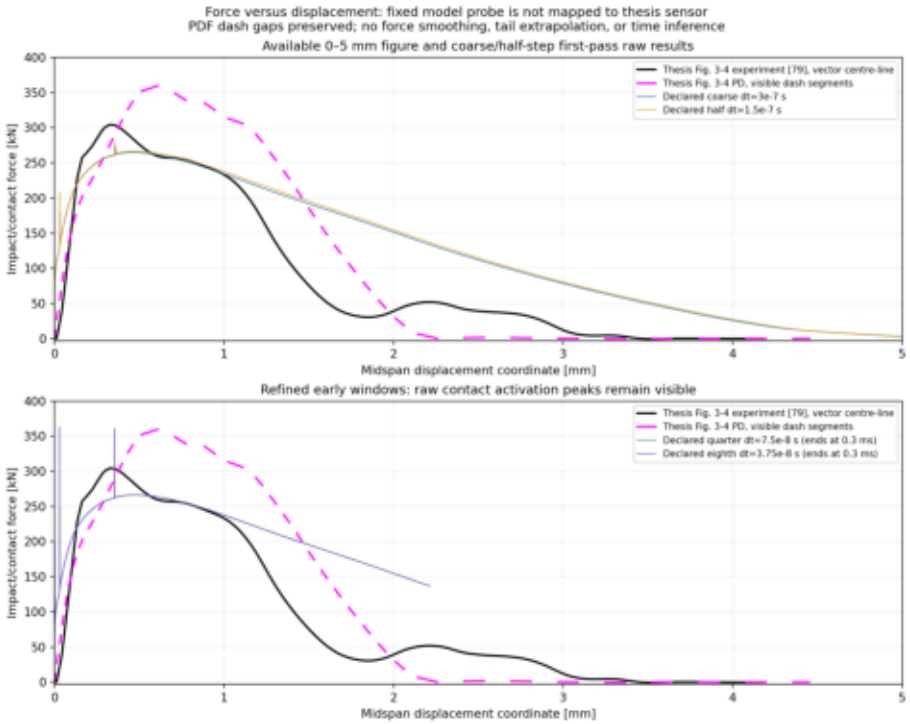


Unfiltered maximum: 272.977271 kN at 0.1416 ms. First positive contact: 0.0099 ms. Integrated impulse through 20.0001 ms: 147.000887 N·s.

Fixed bottom-layer mean at the endpoint: 253.612860 mm, still moving downward at 12.963251 m/s. Final hammer displacement: 169.893807 mm; hammer velocity: −6.971852 m/s. Contact reaction at the endpoint remains approximately 9.217 kN.

The history passed 22.70 and 23.38 mm without reaching a turning point. Stopping at that passage would be an invalid maximum. The 20 ms endpoint is a declared observation window informed by the supplied separate Wang model; it is not Bowen’s recovered stop criterion.

Bowen Fig. 3-4: force–deflection comparison



Source: thesis p. 32 / PDF p. 49. PDF centre-lines preserve 157 black experimental segments and 37 visible PD dashes. Dash gaps remain open; no smoothing, tail extrapolation or force–time history is invented.

Drawn/recorded curve	Peak [kN]	Deflection at peak [mm]
Figure experiment centre-line	304.525	.329650
Figure Bowen PD visible dashes	359.727	.610228
Current dt .30 µs	272.977	.360764
Current dt .0375 µs activation spike	362.702	.030876

Current probe: −mean(u_z) over 28 fixed full-width bottom particles, at x = 297.5/302.5 mm and z = 2.5 mm. Equivalence to Bowen's extraction or the experimental optical ROI is unknown. Reactions are unfiltered; no alignment or processing parameter was fitted.

The 0–5 mm source plot does not establish the complete 22–23 mm event. Coarse/fine traces end at their first 5 mm crossing; finer runs end at .3 ms / 2.213 mm. The eighth-step near-target spike is at only .0309 mm because additional points activate contact.

Drawn experiment apex: 304.525 kN versus Table 3-1's 301.0 kN. Figure legend cites [79], table/text [78]. Extraction estimates the drawn line, not original physical data or uncertainty. Source identity and sensor limitations prevent a matched-observation validation claim.

Fixed displacement probes and structural detachment

Final checkpoint observable	Coarse 20.0001 ms	Half-dt 3 ms
Bottom mean downward [mm]	253.613	33.218
Bottom downward speed [m/s]	12.963	12.181
Support-connected bottom probes	0/28	28/28
Support-connected concrete	56/33,600	31,748/33,600
Support-connected steel	0/270	270/270
Two steel midspan probes [mm]	119.195, 122.548	34.267, 33.612
Detached concrete nodes	33,544	1,852
Maximum any concrete norm [mm]	2125.642	279.751

The graph uses positive-stiffness concrete bonds, positive-weight PDROD links and steel rods, excluding analytically zero horizon-shell interactions. The bearing constraints remain attached to original material IDs: their vertical motion is fixed but final lateral motion reaches 317.019 mm. This permanent material patch differs from spatial contact with a finite stationary bearing. The author’s bearing/friction/anchor definitions are unavailable.

At the production endpoint 33,544 concrete points and all 270 steel nodes are detached from the bearing graph. Maximum any-particle motion exceeds 2.12 m. This is severe collapse/fragment motion, not a bounded 22.70 mm RC beam deflection. A small momentum residual does not validate this structural outcome.

Experimental DIC may follow a separated shear plug, so detachment alone does not disqualify a sensor. The original measured point/ROI and time are still required. Fixed concrete and steel probes remain explicit; a changing supported subset or a maximum displacement norm is never substituted for the reported midspan sensor.

Direct steel contact is omitted here but included in the supplied Fortran. Verified snapshots at 1.5, 3, 6, 12 and 20.0001 ms contain no steel point penetration or bar-radius overlap with the sphere; the final minimum bar-surface clearance is 24.65 mm. No upper-sphere concrete contact appears in those states. This excludes those mechanisms at the inspected snapshots, not at every intervening step.

Energy and momentum: bookkeeping and limits

Exchange-compensated material-energy residual:
D = KE + stored + deleted rupture + steel plastic − initial KE − signed rate-potential exchange.

Rate-dependent threshold updates exchange potential even with static c. Subtracting that signed exchange is an accounting identity, not a passivity or physical-supply proof. Contact/support projection changes remain in D; additionally subtracting projection kinetic changes gives a separate operator-splitting ledger.

Sampled diagnostic	Coarse ≤20 ms	Half-dt ≤3 ms	Original ≤3 ms
Initial hammer KE [J]	2239.488	2239.488	2239.488
Max |compensated residual| [J]	41.856	5.840	2.809
Max residual / initial KE [%]	1.869	0.261	0.125
Max momentum residual [kg m/s]	1.659e-07	4.633e-09	9.529e-10

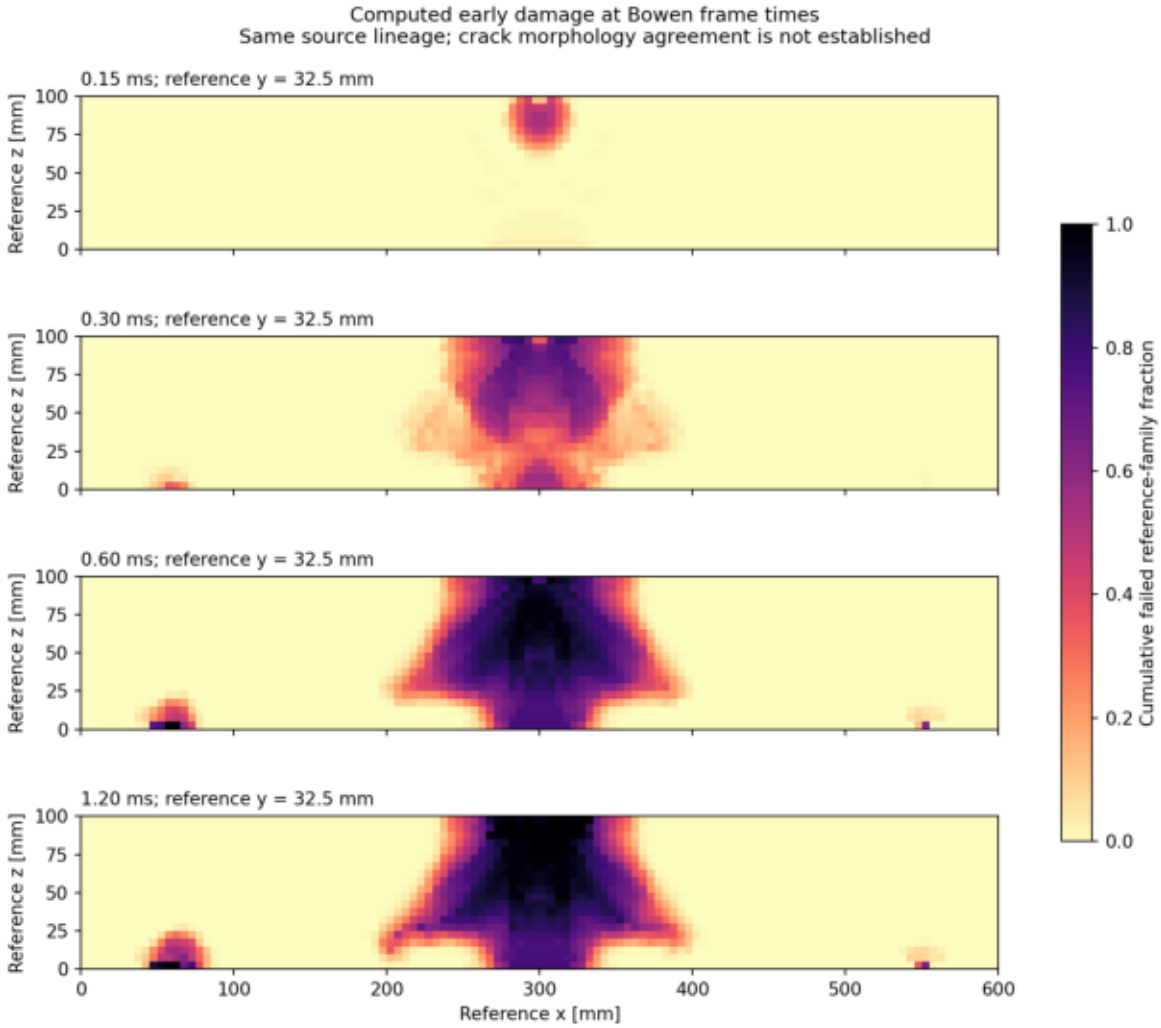
Coarse energy is sampled every 100 steps. Its last scheduled sample is step 66,600 / 19.98 ms; the final step 66,667 is intentionally unsampled, represented by NaN. All 666 scheduled samples and all final numerical state fields are finite. The final summary’s NaN energy field is not evidence of a nonfinite mechanical state.

Coarse energy at 19.98 ms	Value [J]
kinetic J	1801.663170
plastic dissipation J	157.157608
rate potential exchange J	14.491620
rupture deleted energy J	246.646961
stored energy J	6.655395
initial energy J	2239.488000

Exchange-compensated residual at that sample is -41.856485 J; the separate projection-kinetic ledger is 2216.615477 J. The latter omits projection potential changes. Small residual or conserved momentum does not establish correct interface calibration, fracture localization or sensor response.

Independent unchanged-checkpoint evaluation gives accounted energy 2212.106175 J, change -27.381825 J, signed exchange +14.471561 J and compensated residual -41.853385 J (-1.868882%). Stored energy is 7.091248 J; component terms are retained individually.

Crack pattern: reference evidence and acceptance



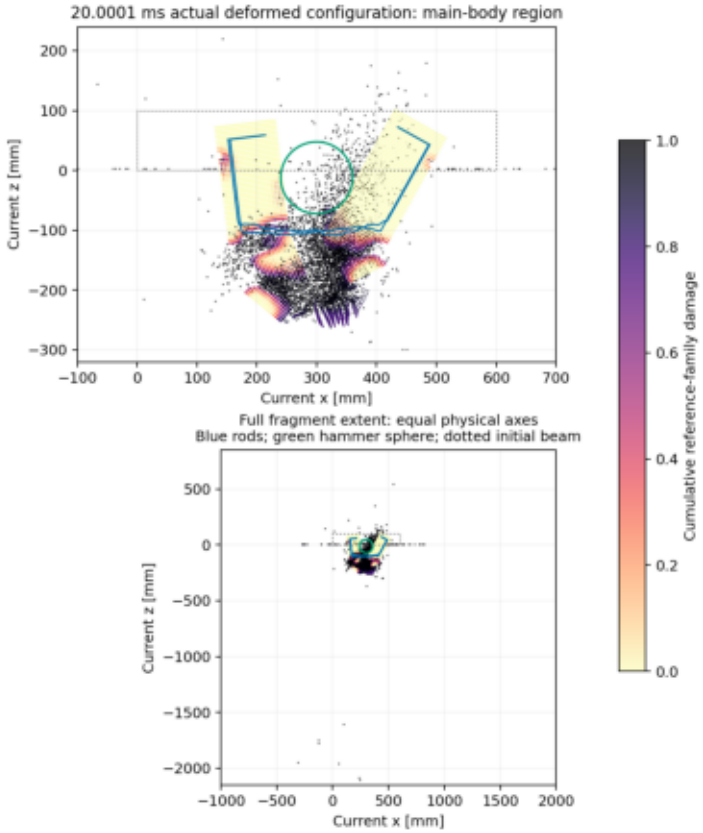
Production lineage at the exact thesis frame times: 0.15, 0.30, 0.60 and 1.20 ms. The fixed central slice uses undeformed reference coordinates, cumulative failed-family volume and a 0–1 scale. The central region becomes highly damaged. Bowen's exact slice/display mapping is unknown; the comparison is qualitative.

Bowen's H=6 m frames show two inclined shear boundaries surrounding a relatively less-damaged plug by 0.30 ms, with crushing above and local bottom spalling. The retained reconstruction at 0.75 ms shows a diffuse, heavily damaged central block; the required plug morphology has not been reproduced.

The thesis damage sequence is at 0.15, 0.30, 0.60 and 1.20 ms. "Damage evolution ended" around 1.20 ms does not mean displacement stopped. Reference-family damage, present mechanical activity, crack opening, connectivity and fragment position must remain separate diagnostics.

Bowen's 0.1701 scalar has an unmatched run/time definition. Production damage is 0.174908 at 1.20 ms; half-dt final 3 ms damage is 0.177927. Nearby global values are not morphology agreement. The exact retained early damage arrays and input-state hashes accompany the figure.

Final production damage and failed load path



Coarse checkpoint 66,667 / 20.0001 ms: 2,656 mechanical components, 2,110 isolated concrete points. Mean cumulative family damage is 0.247886; current inactive-family fraction is 0.244636.

Actual coordinates: blue rods, green hammer and dotted initial beam. The main region is enlarged above; full fragment extent is below with equal physical axes. Reference-coordinate damage slices are retained separately.

PDROD active samples at the two midspan steel points fall from 0 and 1 at 3 ms to zero at both by 6 ms. At 20 ms, 96/270 steel nodes lack active samples; 61/268 rods are plastic (maximum 0.30379), without rupture. Wang's 2D steel PD failure thresholds do not supply an ultimate strain for these 1D rods. Earlier graphs are connectivity upper bounds because dynamic thresholds were not saved; the final graph is exact.

The displacement target is exceeded without a maximum, support connectivity is lost and raw force is timestep sensitive. These establish unsuccessful reproduction, but do not uniquely identify which unresolved interface/support/rate input caused collapse.

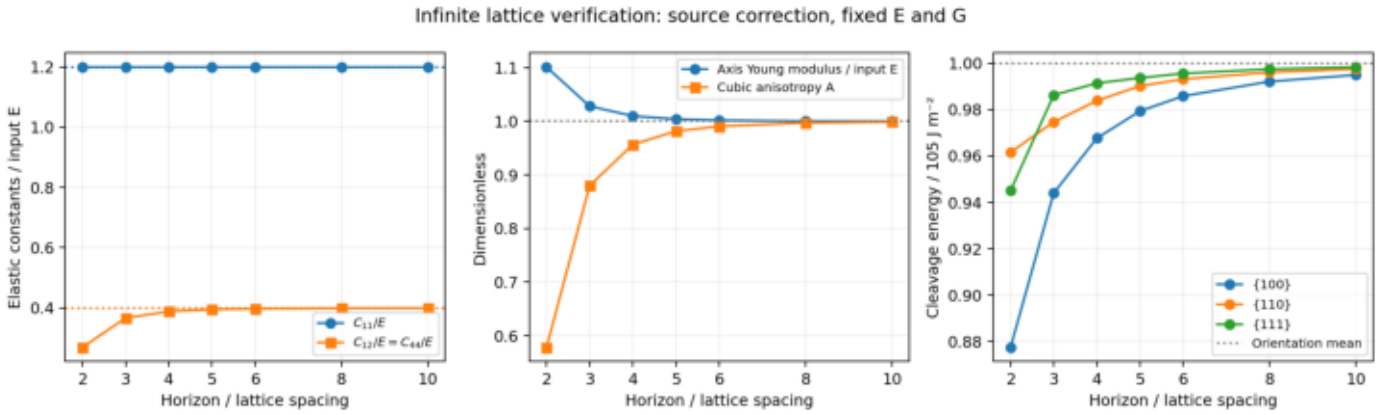
PDROD residuals are below 5×10^{-12} N and 3×10^{-12} N·m. No remaining mass/volume arithmetic error was found. Missing interface calibration and bearing/rate definitions remain essential; strengthening coupling solely to match the target would be unsupported fitting.

Cross-phase geometry: 31 concrete centres lie inside finite 3 mm rod capsules at 3 ms and 61 at 20 ms, versus zero initially. Separate steel–concrete closure contact is absent. The 3,205 restored compressive virtual samples lie outside the physical bar radius; their summed forces are not a bar-contact load. Assigning a 2.5 mm particle collision radius already creates 1,088 initial overlaps, so spacing alone cannot define a valid closure law.

Independent bulk lattice consistency audit

At the production δ/Δx = 3, the lattice is directionally biased despite nominal E = 25 GPa. The energy curvature and independent tensor agree to a maximum relative discrepancy of approximately 1.6×10^{-10}. No impact material input was changed.

δ/Δx	Axis E [GPa]	Axis ν	Cubic A
2	27.52929	0.18318	0.57820
3	25.70301	0.23403	0.87990
4	25.24733	0.24446	0.95661
5	25.10035	0.24768	0.98162
6	25.04801	0.24881	0.99056
8	25.00948	0.24964	0.99715
10	24.99952	0.24986	0.99886



Central bond forces imply C12 = C44. Production C11 = 29.99204 GPa and C12 = C44 = 9.16348 GPa give axis E = 25.70301 GPa, ν = 0.23403 and A = 0.87990. This coarse cubic anisotropy can bias directional crack propagation.

The independent orientation-mean fracture normalization is 105 J/m². At δ/Δx = 3, cleavage energies are 99.1196, 102.3371 and 103.5584 J/m² for {100}, {110}, {111}. The finite source elastic calibration at strain .001 adds a −0.02855% continuum modulus bias.

These are bulk constitutive verification results, not complete impact mesh convergence or justification to alter input E to fit the targets.

Recovered baseline and supplied Fortran evidence

The original reconstructed model completed a retained, independently checked 10,000-step / 3 ms continuation. Its full peak is 220.662764 kN; fixed bottom motion is 38.089413 mm and still downward. None of its 28 bottom probes remains bearing-connected; both steel midspan probes remain connected.

The original runner has no modern completion heartbeat schema. Completion here is established by the retained uninterrupted full history, source-checked aligned final checkpoint and recorded successful calculation, not by absence of a process. The archived baseline is an independent reconstruction, not Stage 19 and not Bowen's original executable.

Baseline checkpoint quantity	At 3 ms
Detached concrete nodes	5,348
Bottom mean / downward velocity	38.089413 mm / 14.703379 m/s
Steel midspan probes	24.076709, 23.046628 mm
Cumulative reference-family damage	0.315812

Console2.f90 is a different case: roughly 3.46 m beam, prescribed two-cylinder motion over 0.3 s, with strong damping. It is not the free 38.4 kg hammer Chapter 3 case.

Its literal 100,000-step run numerically completed with 1059.806437 kN reaction and 1300.168100 m maximum steel displacement. The yielded steel branch copies the axial accumulator into transverse components, producing a verified nonphysical force imbalance.

Five archived corrections address steel component indices, interface return-density control-volume ratio, support velocity reset, steel contact mass and cylinder tangential motion. Seven compiled native operator checks passed; the corrected 200-step elastic run completed. A full corrected 0.3 s native production result is not claimed.

The corrected different-case Fortran still cannot recover missing Chapter 3 material/interface/boundary files. Its original invalid peak must not be used as experimental agreement.

Verification, restart and scientific limits

The modular solver is independent of Stage 13–19 notebooks. Tests cover physical forces and masses, interpolation virtual work, force/torque conservation, rigid motion, rod yielding, contact impulse, force–energy derivatives, rupture deletion and rate-potential exchange.

The recorded full setup execution passed 88 tests in 62.99 s. The focused v4 certificate records 32 passing checks overlapping that setup; it is not an additional aggregate count. Eight lattice checks were run separately. The contact startup BLAS guard checked 4,096 vectors with zero mismatches.

On 2026-10-13, a fresh readiness check verified all seven exact dependency pins and 4,096 contact-arithmetic vectors with zero mismatches. The 88-test whole-suite execution remains its separately dated prior record; it was not rerun for this report.

Atomic checkpoints preserve concrete/steel motion, previous positions, flags and threshold history, rod plastic strain, hammer state and diagnostic accumulators. Restart rejects changed physics/source hashes. Complete CSV prefixes are copied only through the committed checkpoint; absolute target steps prevent silent extension errors.

The accelerated v3 package preserves physical inputs. Coarse and fine checkpoint migrations each replay 100 complete steps with bitwise equality of every saved numeric array and every history row. Separate signal/durability tests verify finishing a committed step after SIGINT/SIGTERM, discarding a partially mutated exception state and preserving a checkpoint-aligned history after forced termination.

Independent validators preserve their exact input CSV/checkpoint bytes, reject incompatible manifests and conflicting restart rows, and separate exchange-compensated material-energy residual from the projection kinetic ledger. The reported production and refinement windows have completed; none establishes the original author’s full model.

Unresolved definitions and validation limits

Complete validation would require source-defined Chapter 3 interface/support inputs, justified contact and observation operators, full impact mesh/time sensitivity and time-matched shear-plug/sensor comparison. Original solver/input and measurement definitions remain missing. The current model is preserved; these gaps do not justify arbitrary fitting.

Continuing ballistic fragment motion solely to reach a desired number is not useful validation. The completed 20 ms calculation is retained as a failed candidate with diagnostics, rather than relabelled as Bowen’s result.

The large Peridynamic articles.zip remains uninspected because the available download tool rejects its size. This report does not claim exhaustive review of every supplied archive.

Dynamic modulus: source interpretation fails passivity

A separate candidate changes only the existing dynamic_stiffness option, from false to true, starting from the undeformed state. Frozen solver source, static material inputs, fracture calibration, rate join, interface, bearings, hammer and timestep are identical. This is a constitutive interpretation, not timestep convergence or a parameter fitted to impact results.

Thesis Eq. 2-14, printed p. 14 / PDF p. 31, calls DIFE the elastic-modulus enhancement factor. Eq. 2-12 is linear in E; Eq. 2-15 scales both onset and terminal stretches by DIFt/DIFE. If those terms have their stated physical meanings, c must also scale by DIFE.

cd = DIFE c; s0d = (DIFt/DIFE) s0
Onset force ratio = (cd/c)(s0d/s0) = DIFt.
With static c, the onset force ratio is DIFt/DIFE.

Rate [1/s]	DIFt	DIFE	Static-c force ratio	Scaled-c force ratio
0	1.00000	1.00000	1.00000	1.00000
1	1.31845	1.22565	1.07571	1.31845
10	2.84051	1.27165	2.23373	2.84051
1000	13.18447	1.36888	9.63156	13.18447

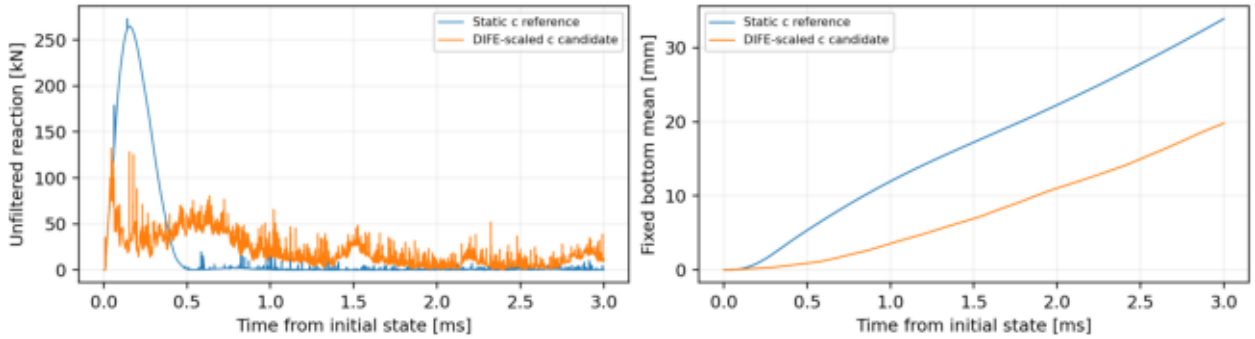
The elementary check calls the actual source-hashed force kernel on a weighted bond with physical nodal volumes. Independent scalar DIF formulas predict each expected strength ratio. All strength, actual-force and rate-potential-exchange checks passed before this impact variant was assessed.

The conclusion is conditional: Appendix A visibly changes thresholds without providing the full micromodulus-update loop and uses different empirical exponents. The original author’s dynamic-c policy is not proved. The continuous-at-1/s rate join and static interface hypothesis remain separate unresolved choices.

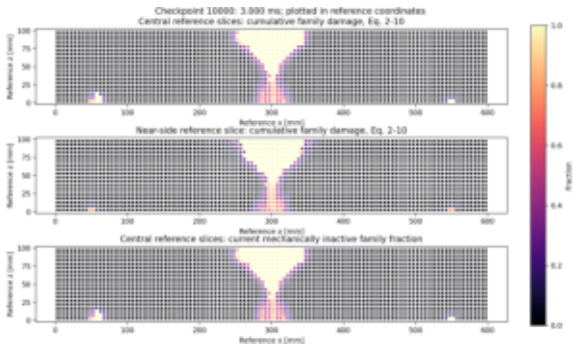
The actual signed instantaneous rate also enhances stiffness when a compressed bond recovers toward zero. An independently prescribed elastic compression/recovery cycle at ±1/s and amplitude 1e−4 returns all force-relevant state to its start yet delivers 0.788217 μJ net mechanical work, 22.5649% of compression input. Static-c control closes; actual-operator refinements preserve this surplus without rupture or plasticity.

This is a demonstrated mechanical-passivity counterexample for this reconstruction, not a proven error in Bowen’s missing solver. Signed rate-potential exchange accounts for the supply algebraically but is not an independently defined physical energy reservoir. Dynamic fracture work scales by DIFt²/DIFE; static 105 J/m² was not retuned. The completed impact case is retained as a diagnostic variant and is not accepted as a validation candidate.

Dynamic modulus: completed 3 ms assessment



Same-time quantity	Static-c reference	DIFE-scaled c
Raw maximum [kN]	272.977271	132.463029
Impulse [N·s]	65.085371	55.122445
Bottom motion / downward speed [mm / m/s]	33.906 / 12.698	19.789 / 9.337
Last accounted energy [J]	2236.626	2344.466
Accounted energy change from initial [J]	-2.862	+104.978
Last signed rate-potential exchange [J]	+22.720	+153.546



Candidate endpoint: 28/28 fixed bottom probes, 270/270 steel nodes and 31,865/33,600 concrete nodes are bearing-connected. Mean cumulative family damage is 0.078449. All numeric final states and 100 scheduled energy samples are finite.

Accounted energy = KE + stored + deleted rupture + steel plastic; initial energy is 2239.488 J. Its increase is +104.977770 J. Subtracting signed exchange gives the exchange-compensated material-energy residual -48.568465 J, maximum magnitude 48.568465 J. Momentum residual is 7.047e-10 kg m/s; last energy sample is 3 ms.

At 3 ms the fixed probe is still descending at 9.337 m/s; 19.789 mm is not a full-event maximum. Neither sensor equivalence nor time-matched shear-plug agreement is established. The mechanical-passivity failure prevents accepting this instantaneous modulus policy as a closed material model. The unchanged nodal contact operator also retains its activation-spike limitation. No original-author policy has been recovered.

PDROD: isolated source-visible force-history law

This fresh case changes only the steel–concrete PDROD failure gate. Concrete failure stays irreversible and concrete stiffness stays static. Geometry, materials, fracture calibration, rate law, bearing constraints, hammer, timestep, contact and fixed probes match the corrected coarse reference. No input was fitted to reported impact results.

Bowen Appendix C, printed pp. 89–91 / PDF pp. 106–108, computes virtual-point displacement with eight trilinear weights, evaluates the force from current stretch, distributes equal/opposite contributions, and then updates permanent failure flags. The visible complete sample loop does not consult an old failure flag before its next force evaluation. The independent Console2.f90 PDROD loop, lines 758–828, has the same force-before-bookkeeping order.

The supplied Fortran uses a different beam and loading case. Matching this visible loop establishes source-operation fidelity for the declared history policy; it does not recover Bowen’s missing Chapter 3 initialization or prove the author used these inputs.

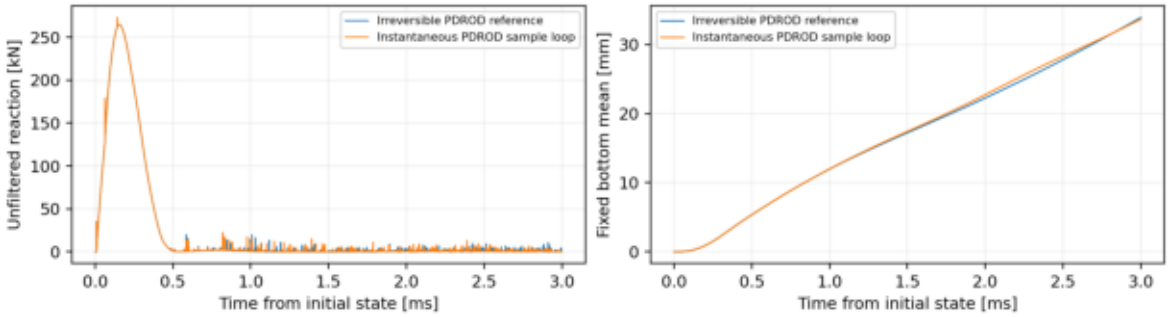
V4 introduces interface_irreversible independently of the concrete gate. With it false, a sample that once exceeded terminal stretch restores compressive force on closure and tensile force if it returns below terminal. The ever-failed flag stays false permanently for damage bookkeeping. A historical damaged sample is therefore not necessarily force-free or disconnected.

The force-law primitive is continuous and capped above terminal stretch. This instantaneous branch has zero irreversible interface rupture-deletion loss. Keeping a permanent rupture loss while recreating the same potential would double-count energy. The production diagnostics use the same independent gate for interface force, potential and mechanical connectivity.

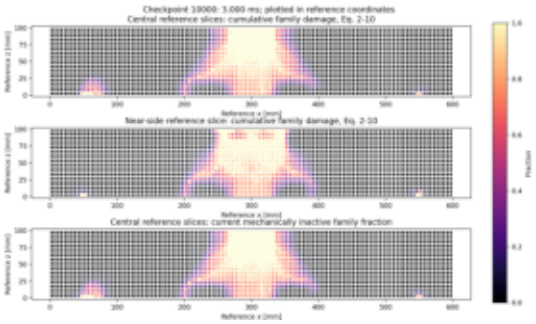
Focused source-pinned checks verify the scalar load-beyond-terminal/compression/tensile-return path, force/torque balance, potential gradients, permanent bookkeeping, independent phase gates, unchanged default-v4/v3 dynamics and checkpoint/restart policy. Only assembly.py and solver.py differ among physical package modules.

Restored tensile bonding is not independently validated adhesive recovery, and this law is not finite-radius bar contact. The 20 mm steel centreline height is dimensioned; hook endpoints and transverse coordinates are not. Bulk interface coefficient/virtual volumes, bearing footprint/lateral constraints, self-contact, steel rupture, rate inputs and experimental sensor remain unresolved. A closer raw peak cannot select this policy as the exact author model.

PDROD: completed 3 ms diagnostic assessment



Same-time quantity	Irreversible reference	Instantaneous PDROD
Raw maximum [kN]	272.977271	272.977271
Contact impulse [N·s]	65.085371	65.026097
Fixed bottom / downward speed [mm / m/s]	33.906 / 12.698	33.668 / 11.514
Accounted energy change [J]	-2.862	-2.790
Signed rate exchange [J]	+22.720	+23.244



At the endpoint 28/28 fixed bottom probes, 270/270 steel and 31,749/33,600 concrete particles are bearing-connected under the active history policy. Mean ever-failed family damage is 0.186067. All numeric final states and 100 scheduled energy samples are finite.

Accounted energy is KE + stored + deleted rupture + steel plastic. Its change and signed exchange are reported separately; exchange-compensated material-energy residual at the last sample is -26.033619 J, maximum magnitude 26.033619 J. Maximum momentum residual is 9.308e-10 kg m/s. Finite states and balanced momentum alone do not validate this interface law.

The fixed probe already exceeds Bowen’s 22.70 mm and remains descending. This fails the claimed maximum-deflection comparison; later integration cannot erase that overshoot. The unchanged nodal impulse-projection contact retains its raw-peak nonconvergence. A credible matching candidate requires independent timestep/mesh verification and full-event assessment before acceptance.

Concrete: isolated source-visible force-history law

This new fresh run changes only the concrete failure gate relative to the completed PDROD-only v4 case. Both phase forces now evaluate current stretch regardless of previous failure, while permanent failure flags still record damage history. Frozen v4 source, static micromodulus, material inputs, geometry, bearings, timestep, hammer, contact, rate law and probes are identical.

The supplied Console2.f90 full concrete loop, lines 611–683, evaluates its current-strain force before updating ever-failed flags, without gating force with an old flag. The reference family built at lines 443–460 remains present. Thesis Table 2-2 likewise orders force before failure bookkeeping; Appendix A only shows rate-threshold updates. These excerpts do not recover the missing full Chapter 3 solver. The Fortran is inspectable source-operation evidence from a different beam/loading case.

Disabling the old concrete failure gate restores compressive force after a bond exceeded terminal tensile stretch. It also restores tensile force if the current stretch returns below terminal. This is not a compression-only closure law or independently identified adhesive healing. Threshold/rate updates remain the declared continuous-at-1/s interpretation; no new force factor or material parameter is introduced.

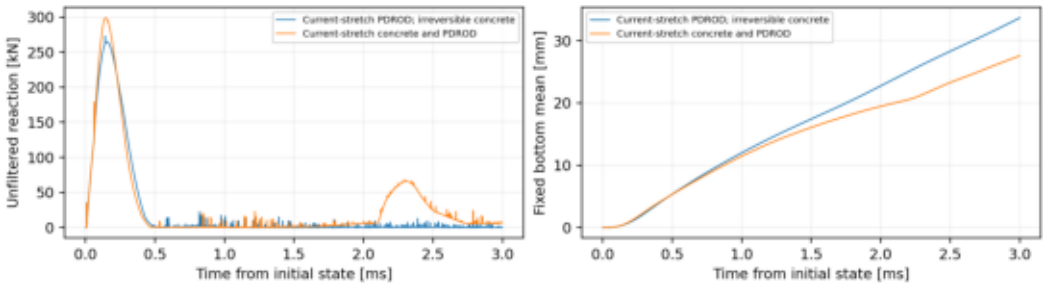
The current-rate bond potential remains the continuous primitive of the instantaneous scalar force and is capped above terminal stretch. Concrete irreversible rupture-deletion loss is zero in this policy, as is PDROD irreversible deletion. Recording irreversible loss while recreating that stored potential would count it twice. Signed rate-potential exchange is retained separately from accounted energy and is not itself a physical reservoir.

Ever-failed family damage and present load-bearing activity therefore differ. A high historical damage value does not mean every flagged bond is currently force-free. Mechanical connectivity must use each phase’s actual current force policy and positive stiffness/interpolation weights; the fixed displacement probe is never replaced by a selected connected subset.

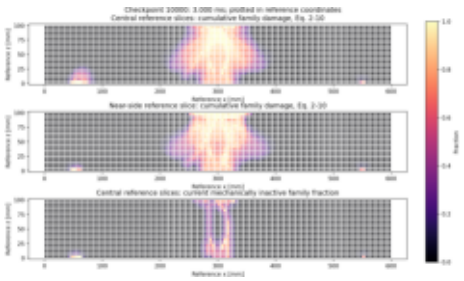
A bounded independent review proves that the configured continuous-1/s signed-rate rule has threshold ratio $q \geq 1$ on loading and $q = 1$ on unloading. Relative to the static force primitive, its extra resisting power is nonnegative. The actual-kernel closed cycle consumes 0.000536586 J external work. This establishes ideal bond-force passivity for these specific choices, not finite-step or full-beam energy conservation, calibrated healing, or author equivalence.

Elementary operator/gradient/bookkeeping checks precede the run. Physical fracture/reversal, source-specific interface calibration, finite bearings, steel rupture and experimental sensor definitions remain unresolved. At the user’s request, the completed model and results are preserved without further parameter changes. Partial scalar agreement is reported explicitly.

Concrete: completed 3 ms diagnostic assessment



Same-time quantity	PDROD-only reference	Both current stretch
Raw maximum [kN]	272.977271	298.951738
Contact impulse [N·s]	65.026097	89.297881
Fixed bottom / downward speed [mm / m/s]	33.668 / 11.514	27.570 / 8.198
Accounted energy change [J]	-2.790	-100.069
Signed rate exchange [J]	+23.244	-44.827



Endpoint connectivity: 28/28 fixed bottom probes, 270/270 steel and 33,573/33,600 concrete particles bearing-connected. Mean ever-failed family damage is 0.161226, distinct from instantaneous activity. All numeric final states and 100 scheduled energy samples are finite.

Experiment comparison: raw peak 298.952 kN versus 301.0 kN (-0.68%); 3 ms fixed motion 27.570 mm versus 23.38 mm (+17.92%). Bowen’s 359.8 kN peak differs by -16.91%. This is close scalar force agreement at this timestep.

Accounted energy is KE + stored + deleted rupture + steel plastic. Both instantaneous phase policies have zero irreversible rupture deletion. Signed exchange is reported separately; exchange-compensated material-energy residual is -55.242201 J, maximum magnitude 55.242201 J. Maximum momentum residual is 1.172e-09 kg m/s. These diagnostics do not certify physical constitutive closure.

The fixed probe exceeds both reported deflection maxima while still descending. This value is a 3 ms observation, not an event maximum or identified experimental-sensor output. Further integration cannot erase the observed overshoot. Time-matched punching/shear crack agreement remains unvalidated. The unchanged nodal contact operator retains its activation-spike limitation. The frozen result has partial experimental agreement; it is not asserted to be Bowen’s exact model.

Source-pinned report provenance

This PDF uses aligned final retained histories/checkpoints, exact package hashes and independent validator metrics. Its companion JSON and snapshot directory retain all 8 run inputs, manifests, CSV prefixes, native checkpoints, figures, completion evidence and scientific audit artifacts.

Production 20 ms: pdbeam_v3; 66,667 steps; checkpoint retained.

Checkpoint SHA256: a1cec9aa7c078260509054fe630ffec4dedaed47221fc5b9d267f4d33c5f2f60

History SHA256: 513d8bb6a3959e703bb7e3f0c8be0a29ead02db0e8d2899a7016d3f22b31ac8d

Half-dt 3 ms: pdbeam_v3; 20,000 steps; checkpoint retained.

Checkpoint SHA256: 2ba567ca45224dfcaf967971c663019d09b598aee3fd3c81c662868bb617c6a5

History SHA256: 9097bbddda147267c7e60e07aa5039a81507142daf2efdd7e94b14fa048ba411

Quarter-dt .3 ms: pdbeam_v3; 4,000 steps; checkpoint retained.

Checkpoint SHA256: ffe42ca84a1c87b758c398fb2854f1dfb4da1db1c0087e6546c3eafed6ad056b

History SHA256: d0d159c1492ece42eb728c9a6a9c23cfb4d6a6869fad91319a1d7dd8379767a3

Eighth-dt .3 ms: pdbeam_v3; 8,000 steps; checkpoint retained.

Checkpoint SHA256: 4a2d15b23697a4bacf435a79402e94cea66ee4721480ab4911fafa7135ef7078

History SHA256: f0f7b6ae87e1d44f1cb47343a82aaa3658489643a64eb8125751f85545204be4

Original 3 ms: pdbeam; 10,000 steps; checkpoint retained.

Checkpoint SHA256: b5f2499edcfc079a06d90d758423474457ad07e5ab95b23dbed24c4fcf778f52

History part 1/2 SHA256: e466fe10312315965f94b1f560a5292235f5a380e66387b41f627874e15ac74c

The original history also requires its archived 1–5,000 prefix. All prefix hashes and complete records are preserved in JSON metadata. A source/configuration hash identifies the implemented scientific operator; it does not establish equivalence to the missing author code.

Source-pinned report provenance continued

This PDF uses aligned final retained histories/checkpoints, exact package hashes and independent validator metrics. Its companion JSON and snapshot directory retain all 8 run inputs, manifests, CSV prefixes, native checkpoints, figures, completion evidence and scientific audit artifacts.

Dynamic modulus candidate: pdbeam_v3; 10,000 steps; checkpoint retained.

Checkpoint SHA256: 8aa50da5261c4a534b5be627a78bced2b6ddcc400568262a1d7160a153c123b7

History SHA256: 8ee9a8b89b64e4f33a6c1719873dc64f0f559ab32ef4d0642c0bc69157c8fb3d

Instantaneous PDROD candidate: pdbeam_v4; 10,000 steps; checkpoint retained.

Checkpoint SHA256: f3f6be8e15d28660912f0106f4cd518749f4a2289546c10e9a082de98e3219d7

History SHA256: ff54494d2da2a8bd5a7e629b2704ee6f22633d3b3cc0165b82fdb7312c3ff71a

Instantaneous concrete candidate: pdbeam_v4; 10,000 steps; checkpoint retained.

Checkpoint SHA256: 6197e9e7461d2a26254ffd919407b77a4e1086bf35b4466da91dbf1356b180c3

History SHA256: 775a864f3a60be1c4a083a987e865c5620304f91814a931376eaa7ff02ac1cbb

The original history also requires its archived 1–5,000 prefix. All prefix hashes and complete records are preserved in JSON metadata. A source/configuration hash identifies the implemented scientific operator; it does not establish equivalence to the missing author code.

Run completion observations at report time

corrected_20ms_accelerated: completed requested window.

corrected_halfdt_3ms: completed requested window.

corrected_quarterdt_0p3ms: completed requested window.

corrected_eighthdt_0p3ms: completed requested window.

recovered_original_3ms: completed 3 ms window (retained history/checkpoint evidence).

dynamic_modulus_3ms: completed requested window.

instantaneous_pdrod_3ms: completed requested window.

instantaneous_concrete_pdrod_3ms: completed requested window.

Rebuild using research/reporting/README.md with the optional validation paths recorded in companion JSON.

The generator refuses changed source/evidence bytes, writes atomically, and does not alter solver code, raw results or GitHub. Public PDF publication is a separate reviewed action.